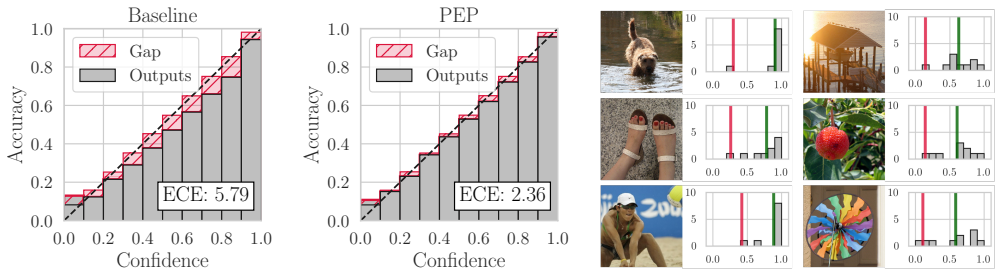
Table 1: ImageNet results: For all models except VGG19, PEP achieves statistically significant improvements in calibration compared to baseline (BL) and temperature scaling (TS), in terms of NLL and Brier score. PEP also reduces test errors, while TS does not have any effect on test errors. Although TS and PEP outperform baseline in terms of ECE% for DenseNet121, DenseNet169, ResNet, and VGG16, the improvements in ECE% is not consistent among the methods. T^* and σ^* denote optimized temperature for TS and optimized sigma for PEP, respectively. Boldfaced font indicates the best results for each metric of a model and shows that the differences are statistically significant (p-value<0.05).

Model	T^*	σ^* $\times 10^{-3}$	Negative log-likelihood			Brier score			ECE%			Top-1 error %	
			BL	TS	PEP	BL	TS	PEP	BL	TS	PEP	BL	PEP
DenseNet121	1.10	1.94	1.030	1.018	**0.997**	0.357	0.356	**0.349**	3.47	**1.52**	2.03	25.73	**25.13**
DenseNet169	1.23	2.90	1.035	1.007	**0.940**	0.354	0.350	**0.331**	5.47	**1.75**	2.35	25.31	**23.74**
IncepttionV3	0.91	1.94	0.994	0.975	**0.950**	0.328	0.328	**0.317**	**1.80**	4.19	2.46	22.96	**22.26**
ResNet50	1.19	2.60	1.084	1.057	**1.023**	0.365	0.362	**0.350**	5.08	**1.97**	2.94	26.09	**25.18**
VGG16	1.09	1.84	1.199	1.193	**1.164**	0.399	0.399	**0.391**	2.52	2.08	**1.64**	29.39	**28.83**
VGG19	1.09	1.03	1.176	1.171	1.165	0.394	0.394	0.391	4.77	4.50	4.48	28.99	28.75



(a) (b) (c)

Figure 2: Improving pre-trained DenseNet169 with PEP (M=10). (a) and (b) show the reliability diagrams of the baseline and the PEP. (c) shows examples of misclassifications corrected by PEP. The examples were among those with the largest PEP effect on the correct class probability. (c) Top row: brown bear and lampshade changed into Irish terrier and boathouse; Middle row: band aid and pomegranate changed into sandal and strawberry; Bottom row: bathing cap and wall clock changed into volleyball and pinwheel. The histograms at the right of each image illustrate the probability distribution of ensemble. Vertical red and green lines show the predicted class probabilities of the baseline and the PEP for the correct class label. (For more reliability diagrams see Supplementary Material.)

3.1 ImageNet experiments

We evaluated the performance of PEP using large scale networks that were trained on ImageNet (ILSVRC2012) [40] dataset. We used the subset of 50,000 validation images and labels that is included in the development kit of ILSVRC2012. From the 50,000 images, 5,000 images were used as a validation set for optimizing σ in PEP, and temperature T in temperature scaling. The remaining 45,000 images were used as the test set. Golden section search [35] was used to find the σ^* that maximizes $\mathbb{L}(\sigma)$. The search range for σ was 5×10^{-5}–5×10^{-3}, ensemble size was 5 ($m = 5$), and number of iterations was 7. On the test set with 45,000 images, PEP was evaluated using σ^* and with ensemble size of 10 ($m = 10$). Single crops of the center of images were used for the experiments. Evaluation was performed on six pre-trained networks from the Keras library[4]: DenseNet121, DenseNet169 [17], InceptionV3 [44], ResNet50 [13], VGG16, and VGG19 [43]. For all pre-trained networks, Gaussian perturbations were added to the weights of all convolutional layers. Table 1 summarizes the optimized T and σ values, model calibration in terms of NLL, Brier score, and classification errors. For all the pre-trained networks, except VGG19, PEP achieves statistically significant improvements in calibration compared to the baseline and temperature scaling. Note the reduction in top-1 error of DenseNet169 by about 1.5 percentage points, and the reduction in all top-1 errors. Figure 2 shows the reliability diagram for DenseNet169, before and after calibration with PEP with some corrected misclassification examples.

Table 2: MNIST, Fashion MNIST, CIFAR-10, and CIFAR-100 results. The table summarizes experiments described in Section 3.2.

Experiment	Baseline	PEP	Temp. Scaling	MCD	SWA	Deep Ensembles
			NLL			
MNIST (MLP)	0.096 ± 0.01	0.079 ± 0.01	0.074 ± 0.01	0.094 ± 0.00	0.067 ± 0.00	0.044 ± 0.00
MNIST (CNN)	0.036 ± 0.00	0.034 ± 0.00	0.032 ± 0.00	0.031 ± 0.00	0.028 ± 0.00	0.021 ± 0.00
Fashion MNIST	0.360 ± 0.01	0.275 ± 0.01	0.271 ± 0.01	0.218 ± 0.01	0.277 ± 0.01	0.198 ± 0.00
CIFAR-10	1.063 ± 0.03	0.982 ± 0.02	0.956 ± 0.02	0.798 ± 0.01	0.827 ± 0.01	0.709 ± 0.00
CIFAR-100	2.685 ± 0.03	2.651 ± 0.03	2.606 ± 0.03	2.435 ± 0.03	2.314 ± 0.02	2.159 ± 0.01
			Brier			
MNIST (MLP)	0.037 ± 0.00	0.035 ± 0.00	0.035 ± 0.00	0.040 ± 0.00	0.032 ± 0.00	0.020 ± 0.00
MNIST (CNN)	0.016 ± 0.00	0.015 ± 0.00	0.015 ± 0.00	0.014 ± 0.00	0.013 ± 0.00	0.010 ± 0.00
Fashion MNIST	0.137 ± 0.01	0.127 ± 0.01	0.126 ± 0.00	0.111 ± 0.00	0.121 ± 0.00	0.096 ± 0.00
CIFAR-10	0.469 ± 0.01	0.450 ± 0.01	0.447 ± 0.01	0.381 ± 0.01	0.373 ± 0.00	0.335 ± 0.00
CIFAR-100	0.795 ± 0.01	0.786 ± 0.01	0.782 ± 0.01	0.768 ± 0.01	0.723 ± 0.00	0.695 ± 0.00
			ECE %			
MNIST (MLP)	1.324 ± 0.16	0.528 ± 0.12	0.415 ± 0.10	2.569 ± 0.17	0.536 ± 0.08	0.839 ± 0.08
MNIST (CNN)	0.517 ± 0.07	0.366 ± 0.08	0.259 ± 0.06	0.832 ± 0.06	0.282 ± 0.04	0.287 ± 0.05
Fashion MNIST	5.269 ± 0.22	1.784 ± 0.54	1.098 ± 0.18	1.466 ± 0.30	3.988 ± 0.11	0.942 ± 0.13
CIFAR-10	11.718 ± 0.72	4.599 ± 0.82	1.318 ± 0.26	7.109 ± 0.62	8.655 ± 0.29	8.867 ± 0.23
CIFAR-100	9.780 ± 0.69	5.535 ± 0.50	2.012 ± 0.31	12.608 ± 0.59	7.180 ± 0.48	11.954 ± 0.29
			Classification Error %			
MNIST (MLP)	2.264 ± 0.22	2.286 ± 0.24	2.264 ± 0.22	2.452 ± 0.14	2.082 ± 0.10	1.285 ± 0.05
MNIST (CNN)	0.990 ± 0.13	0.990 ± 0.12	0.990 ± 0.13	0.842 ± 0.06	0.868 ± 0.06	0.659 ± 0.03
Fashion MNIST	8.420 ± 0.32	8.522 ± 0.34	8.420 ± 0.32	7.692 ± 0.34	7.734 ± 0.11	6.508 ± 0.10
CIFAR-10	33.023 ± 0.68	32.949 ± 0.74	33.023 ± 0.68	27.207 ± 0.66	26.004 ± 0.36	22.880 ± 0.21
CIFAR-100	64.843 ± 0.69	64.789 ± 0.69	64.843 ± 0.69	60.772 ± 0.58	58.092 ± 0.42	53.917 ± 0.30

3.2 MNIST and CIFAR experiments

The MNIST handwritten digits [27] and fashion MNIST [47] datasets consist of 60,000 training images and 10,000 test images. The CIFAR-10 and CIFAR-100 datasets [24] consists of 50,000 training images and 10,000 test images. We created validation sets by setting aside 10,000 and 5,000 training images from MNIST (handwritten and fashion) and CIFAR, respectively. For the handwritten MNIST dataset, the predictive uncertainty was evaluated for two different neural networks: a Multi-layer Perception (MLP) and a Convolutional Neural Network (CNN) similar to LeNet [29] but with smaller kernel sizes. The MLP is similar to the one used in [26] and has 3 hidden layers with 200 neurons each, ReLu non-linearities, and BN after each layer. For MCD experiments, dropout layers were added after each hidden layer with 0.5 dropout rate as was suggested in [8]. The CNN for MNIST (handwritten and fashion) experiments has two convolutional layers with 32 and 64 kernels of sizes 3×3 with stride size of 1 followed by two fully connected layers (with 128 and 64 neurons each) with BN after both types of layers. Here, again for MCD experiments, dropout was added after all layers with 0.5 dropout rate, except the first and last layers. For the CIFAR-10 and CIFAR-100 dataset, the CNN architecture has 2 convolutional layers with 16 kernels of size 3×3 followed by a max-pooling of 2×2; another 2 convolutional layers with 32 kernels of size 3×3 followed by a max-pooling of size 2×2 and a dense layers of size 128, and finally, a dense layer of 10 for CIFAR-10 and 100 for CIFAR-100. BN was applied to all convolutional layers. For MCD experiments, dropout was added similar to CNN for MNIST experiments. Each network was trained and evaluated 25 times with different initializations of parameters (weights and biases) and random shuffling of the training data. For optimization, stochastic gradient descent with the Adam update rule [22] was used. Each baseline was trained for 15 epochs. The training was carried out for another 25 rounds with dropout for MCD experiments. Models trained and evaluated with active dropout layers were used for MCD evaluation only, and baselines without dropout were used for the rest of the experiments. The Deep Ensembles method was tested by averaging the output of the 10 baseline models. MCD was tested on 25 models and the performance was averaged over all 25 models. Temperature scaling and PEP were tested on the 25 trained baseline models without dropout and the results were averaged.

Table 2 compares the calibration quality and test errors of baselines, PEP, temperature scaling [12], MCD [8], Stochastic Weight Averaging (SWA) [19], and Deep Ensembles [26]. The averages and standard deviation values for NLL, Brier score, and ECE% are provided. For all cases, it can be seen that PEP achieves better calibration in terms of lower NLL compared to the baseline. Deep Ensembles

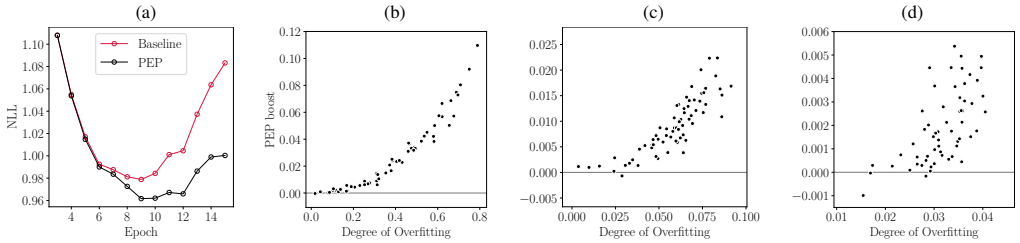


Figure 3: The relationship between overfitting and PEP effect. (a) shows the average of NLLs on test set for CIFAR-10 baselines (red line) and PEP $\mathbb{L}$ (black line). The baseline curve shows overfitting as a result of overtraining. The degree of overfitting was calculated by subtracting the training NLL (loss) from the test NLL (loss). PEP reduces the effect of overfitting and improves log-likelihood. The PEP effect is more substantial as the overfitting grows. (b), (c), and (d) show scatter plots of overfitting vs PEP effect for CIFAR-10, MNIST(MLP), and MNIST(CNN), respectively.

achieves the best NLL and classification errors in all the experiments. Compared to the baseline, temperature scaling and MCD improve calibration in terms of NLL for all three experiments.

Non-Gaussian Distributions We performed limited experiments to test the effect of of using non-Gaussian distributions. We tried perturbing by a uniform distribution with MNIST (MLP) and observed similar performance to a normal distribution. Further tests with additional benchmarks and architectures are needed for conclusive findings.

Effect of Overfitting on PEP effect We ran experiments to quantify the effect of overfitting on PEP effect, and optimized σ values. For the MNIST and CIFAR-10 experiments, model checkpoints were saved at the end of each epoch. Different levels of overfitting as a result of over-training were observed for the three experiments. σ^* was calculated for each epoch and PEP was performed and the PEP effect was measured. Figure 3 (a), shows the effect of calibration and reducing NLL for CIFAR-10 models. Figures 3 (b-d) shows that PEP effect increases with overfitting. Furthermore, we observed that the σ^* values also increase with overfitting, meaning that larger perturbations are required for more overfitting.

Out-of-distribution detection We performed experiments similar to Maddox et al. [32] for out-of-distribution detection. We trained a WideResNet-28x10 on data from five classes of the CIFAR-10 dataset and then evaluated on the whole test set. We measured the symmetrized Kullback–Leibler divergence (KLD) between the in-distribution and out-of-distributions samples. The results show that using PEP, KLD increased from 0.47 (baseline) to 0.72. In the same experiment temperature scaling increased KLD to 0.71.

4 Conclusion

We proposed PEP for improving calibration and performance in deep learning. PEP is computationally inexpensive and can be applied to any pre-trained network. On classification problems, we show that PEP effectively improves probabilistic predictions in terms of log-likelihood, Brier score, and expected calibration error. It also nearly always provides small improvements in accuracy for pre-trained ImageNet networks. We observe that the optimal size of perturbation and the log-likelihood increase from the ensemble correlates with the amount of overfitting. Finally, PEP can be used as a tool to investigate the curvature properties of the likelihood landscape.

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